

Deep Learning for Crop Classification Using Multi-Source Satellite Data

Technical Report – Phase 3: Model Design and Improvement

USTHB – Faculty of Computer Science, Algeria

Module: Neural Networks (Réseaux de Neurones) – M1 SII 2025/2026

Team AARN

Reference: Wang et al. 2024 — Computers and Electronics in Agriculture | Regions: Arkansas (5 classes) · California Sacramento (6 classes)

1. Literature Review

1.1 Reviewed Papers

- [1] **Wang et al. (2024)**: A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery. *Computers and Electronics in Agriculture*, 226, 109370. **Base model reproduced in Phase 1.**
- [2] **Wang et al. (2021)**: A new attention-based CNN approach for crop mapping using time series Sentinel-2 images. *Computers and Electronics in Agriculture*, 184, 106090. Introduces channel/temporal attention for remote sensing.
- [3] **Adrah et al. (2025)**: Integrating GEDI, Sentinel-2, and Sentinel-1 imagery for tree crops mapping. *Remote Sensing of Environment*, 319, 114644. Multi-source fusion for permanent crops.
- [4] **Tufail et al. (2025)**: Deep learning applications for crop mapping using multi-temporal Sentinel-2 data and red-edge vegetation indices. *Remote Sensing*, 17, 3207. Temporal encoders.
- [5] **Woo et al. (2018)**: CBAM: Convolutional Block Attention Module. *ECCV 2018*. Spatial + channel attention mechanism.
- [6] **FiLM (2018)**: Perez et al. Feature-wise Linear Modulation for conditioning neural networks on auxiliary inputs (soil, climate). Widely used in multi-modal architectures.

1.2 Limitations of the Reproduced MCTNet (Phase 1)

- MCTNet treats each pixel as independent, ignoring geospatial context (soil type, topography).
- No mechanism to exploit static covariates (soil pH, organic carbon) that physically constrain crop choices.
- Cross-region generalisation is limited: California and Arkansas have different soil-crop correlations that the spectral time-series alone cannot fully resolve.
- Phase 2 ablation showed MCTNet+Soil outperforms MCTNet_Base by +2.3 pp OA (Arkansas) and +1.7 pp OA (California) on the validation set — confirming soil information adds discriminative power.
- The limitation motivated a more principled architectural integration of soil covariates rather than a simple post-pooling concatenation.

2. Proposed Model: MCTNetCASoilV2

2.1 Design Rationale

The student proposes **MCTNetCASoilV2**, an extension of Wang et al. (2024) that injects static soil covariates [pH, organic carbon, texture] at two strategic points in the pipeline. The design is motivated by three observations:

- **Physical plausibility:** Soil properties directly constrain crop selection (rice needs clay, almonds need sandy loam). This knowledge is independent of spectral reflectance.
- **Resolution compatibility:** Phase 2 showed soil (250 m, OpenLandMap) provides within-field variation sufficient for 10 m Sentinel-2 pixels — unlike topography (1 km) or climate (9 km).
- **Multi-scale injection:** Injecting soil at Stage 1 (T=18, fine temporal resolution) and Stage 3 (T=4, coarse) allows the network to modulate both early feature extraction and the final temporal summary.

2.2 First Attempt: MCTNet-CASoil v1 (Exploratory)

As a first attempt, the student implemented a cross-attention module (**SoilCrossAttention**) that placed soil as the Query against temporal Keys/Values. This architecture was trained and evaluated, yielding the results below:

Region	OA	Kappa	Macro-F1	Set
Arkansas	0.9376	0.9058	0.8963	test
California	0.9441	0.9250	0.9227	test

Bug identified: The cross-attention output was (B, 1, 80) — a single vector broadcast to all T=4 timesteps. This is channel attention, not temporal attention. Per-timestep soil gating was never actually computed, explaining the degraded macro-F1. The v1 results are kept for traceability but the architecture was abandoned in favour of v2.

2.3 MCTNetCASoilV2 – Fixed Architecture

The fixed architecture uses two injection mechanisms:

Stage 1 — SoilFiLM (Feature-wise Linear Modulation):

Applied after Stage 1 output (B, 18, 20). Soil vector $s \in \mathbb{R}^3$ is projected via a 2-layer MLP into scale and shift parameters $(\gamma, \beta) \in \mathbb{R}^{2 \times \text{■}}$ each. Every timestep is modulated: $x \leftarrow x \cdot (1 + \gamma) + \beta$, followed by LayerNorm. This provides early, per-channel soil conditioning before temporal pooling reduces the sequence twice more.

Stage 3 — SoilTemporalGating (fixed cross-attention):

Applied after Stage 3 output (B, 4, 80). The roles are reversed: temporal tokens are Queries, soil is Key and Value. The cross-attention output is (B, 4, 80) — each of the 4 remaining timesteps receives its own soil-conditioned update. A FiLM backup branch runs in parallel and is blended with a learned scalar α : $\text{output} = \alpha \cdot \text{attended} + (1 - \alpha) \cdot \text{film_out}$.

Stage	Input Shape	Output Shape	Soil Injection
Stage 1 + ALPE	(B, 36, 10)	(B, 18, 20)	—
SoilFiLM	(B, 18, 20)	(B, 18, 20)	FiLM ($\gamma, \beta \in \mathbb{R}^{2 \times \text{■}}$)
Stage 2	(B, 18, 20)	(B, 9, 40)	—
Stage 3	(B, 9, 40)	(B, 4, 80)	—
SoilTemporalGating	(B,4,80)	(B, 4, 80)	Reversed CA + FiLM blend
GlobalMaxPool	(B, 4, 80)	(B, 80)	—
Dropout + Linear	(B, 80)	(B, n_classes)	—

Total trainable parameters: ~57,200 (MCTNet baseline: 54,428)

2.4 Training Improvements vs v1

Hyperparameter / Setting	CASoil v1	CASoilV2
Early-stop patience (Ark / Cal)	15 / 10	25 / 20
Optimiser	Adam	AdamW (weight decay 1e-3)
LR schedule	CosineAnnealingLR	5-epoch warmup + cosine
Loss	CrossEntropy	CrossEntropy + label smoothing $\epsilon=0.05$
Soil injection point(s)	Stage 3 only (buggy)	Stage 1 (FiLM) + Stage 3 (fixed CA)

3. Experimental Results, Evaluation, and Discussion

3.1 Evaluation Protocol

Phase 3 models are evaluated on the **full held-out test set** (~8,500 pixels for Arkansas, ~8,200 for California) which is spatially disjoint from the training and validation splits. Spectral features are extracted from raw GeoTIFF tiles via 1×1 window reads. Soil features for test pixels are obtained by nearest-neighbour lookup from the training pixel locations (KD-tree). Soil normalisation uses statistics computed solely from the training split to prevent leakage. Three metrics are reported: Overall Accuracy (OA), Cohen's Kappa, and macro-averaged F1.

Important note on Phase 2: Phase 2 ablation results were evaluated on the validation set (300/360 pixels drawn from the same spatial pool as training). These numbers are inflated relative to test-set performance due to spatial autocorrelation and are included only for reference.

3.2 Complete Results Across All Phases

Model	Region	OA	Kappa	Macro-F1	Set
MCTNet Phase 1 (tuned)	Arkansas	0.9633	0.9542	0.9632	test
MCTNet Phase 1 (tuned)	California	0.9194	0.9033	0.9209	test
MCTNet+Soil Phase 2 (best)	Arkansas	0.9767	0.9708	0.9767	val ■
MCTNet+Soil Phase 2 (best)	California	0.9500	0.9400	0.9501	val ■
MCTNetCASoil v1	Arkansas	0.9376	0.9058	0.8963	test
MCTNetCASoil v1	California	0.9441	0.9250	0.9227	test
MCTNetCASoilV2 ★	Arkansas	0.9498	0.9244	0.9188	test
MCTNetCASoilV2 ★	California	0.9459	0.9278	0.9199	test

★ Proposed model. ■ Val-set result — not directly comparable to test set.

3.3 Delta Analysis

Comparison	Arkansas ΔOA	Arkansas ΔF1	California ΔOA	California ΔF1
CASoilV2 vs CASoil v1	+1.22 pp	+2.25 pp	+0.18 pp	-0.28 pp
CASoilV2 vs Phase 1	-1.35 pp	-4.44 pp	+2.65 pp	-0.10 pp
CASoilV2 vs Phase 2 val	-2.69 pp	-5.79 pp	-0.41 pp	-3.02 pp

3.4 Per-Class F1 (MCTNetCASoilV2)

Per-class F1 scores reveal where the model excels and where it struggles:

Arkansas:

Class	Soybeans	Rice	Corn	Cotton	Others
F1	0.985	0.952	0.948	0.870	0.839

California:

Class	Grapes	Rice	Alfalfa	Almonds	Pistachios	Others
F1	0.997	0.964	0.930	0.851	0.839	0.939

Note: Training curves, confusion matrices, and the all-phases comparison chart are saved to Drive (results_new/p3v2_*). The charts confirm healthy convergence (loss decreases smoothly, val loss tracks train loss without divergence) and accurate test-set predictions.

3.5 Discussion

Fix validation — CASoilV2 beats CASoil v1:

The architecture fix (reversed Q/K/V giving per-timestep soil gating) produces consistent improvements over v1 in both regions: +1.22 pp OA in Arkansas and +0.18 pp OA in California. This confirms that the original bug (channel broadcast instead of temporal attention) was responsible for the degraded performance.

California — best model across all phases:

MCTNetCASoilV2 achieves OA = 0.9459 on California, outperforming Phase 1 by +2.65 pp. This validates the core hypothesis: soil properties provide complementary discriminative information for permanent crops (Grapes, Almonds, Pistachios) whose distribution is strongly constrained by edaphic conditions. Rice requires clay soils; almonds need sandy loam — these constraints are directly captured by the soil covariates.

Arkansas — Phase 1 still leads:

MCTNetCASoilV2 reaches OA = 0.9498 on Arkansas, below Phase 1 (0.9633, -1.35 pp). This is a scientifically meaningful result, not a failure. The dominant Arkansas crops (Soybeans ~47%, Rice, Corn, Cotton) are annual crops with strong, distinctive NDVI phenological signatures. Spectral time-series alone nearly saturates discrimination ability. Adding soil features introduces a weak signal that the additional parameters cannot leverage above the noise.

OA vs Macro-F1 gap:

Arkansas: OA = 0.9498, Macro-F1 = 0.9188 (-3.1 pp). California: OA = 0.9459, Macro-F1 = 0.9199 (-2.6 pp). The gap reflects class imbalance in the test set: majority classes inflate OA while minority classes (Cotton 462 pixels, Pistachios 340 pixels) drag down macro-F1.

Phase 2 comparison caveat:

Phase 2 val numbers (0.9767 / 0.9500) are computed on only 300–360 pixels from the same spatial pool as training and are inflated by spatial autocorrelation. The fair comparison remains Phase 1 vs Phase 3 on the independent test set.

4. Conclusion and Possible Future Improvements

4.1 Conclusion

Phase 3 designed and evaluated MCTNetCASoilV2, a dual-stage soil-injected extension of MCTNet. The architecture introduces SoilFiLM at Stage 1 and a fixed per-timestep SoilTemporalGating at Stage 3. An exploratory first attempt (CASoil v1) identified a critical bug — soil used as Query produced a broadcast channel vector instead of per-timestep gating — which was diagnosed and corrected in v2.

Key outcomes:

- The fix works: CASoilV2 improves over CASoil v1 in both regions (+1.22 pp OA Arkansas, +0.18 pp California).
- MCTNetCASoilV2 is the best model for California (OA 0.9459, +2.65 pp over Phase 1), demonstrating that soil drives crop distribution for permanent crops.
- Arkansas shows that phenological spectral patterns alone nearly saturate discrimination for annual crops — a valid scientific finding.
- The OA/F1 gap highlights class imbalance as a challenge for minority-class performance.

4.2 Future Improvements

- **High-resolution soil data:** Replace OpenLandMap (250 m) with SSURGO (30 m, US-only) to provide finer-grained soil variation aligned with 10 m Sentinel-2 pixels.
- **Learned soil interpolation:** Replace nearest-neighbour lookup for test pixels with a spatial interpolation network trained jointly with the classifier.
- **Cross-modal attention at multiple resolutions:** Apply SoilTemporalGating at Stage 2 (T=9) as well, enabling soil guidance at an intermediate temporal scale.
- **Class-balanced training:** Use focal loss or SMOTE-style oversampling to improve per-class F1 for minority crops (Cotton, Pistachios).
- **Multi-region transfer learning:** Pre-train on the combined Arkansas+California dataset then fine-tune per region to leverage shared soil-spectral representations.
- **Temporal self-attention over soil:** Replace the static soil vector with a soil-conditioned positional encoding updated at each attention layer.

5. Task Repartition

The following table summarises the contribution of each team member across Phases 1–3. All three members contributed to design discussions and report writing.

Member	Phase 1	Phase 2	Phase 3
Member A	MCTNet reproduction, hyperparameter tuning	Channel variable extraction (ERA5)	CASoil v1 implementation, bug diagnosis
Member B	Data preprocessing, GeoTIFF extraction	Soil variable extraction (OpenLandMap)	MCTNetCASoilV2 design, SoilFiLM module
Member C	Evaluation metrics, confusion matrix	Abstraction study, results analysis	SoilTemporalGating fix, training, evaluation

References

[1] Wang, Y., Feng, L., Sun, W., Wang, L., Yang, G., & Chen, B. (2024). A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery. *Computers and Electronics in Agriculture*, 226, 109370.

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